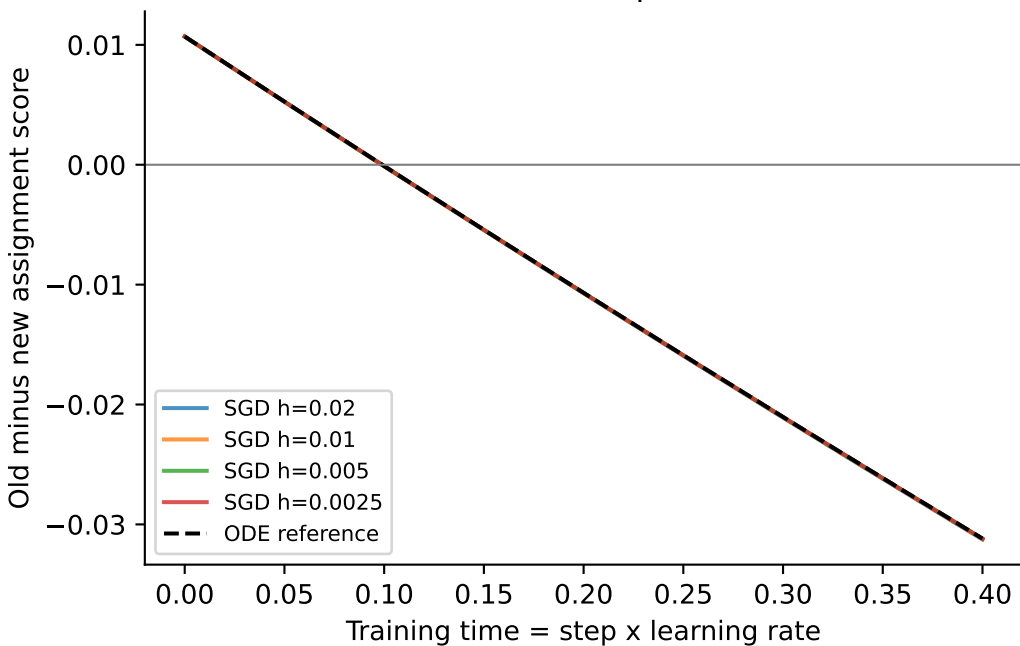
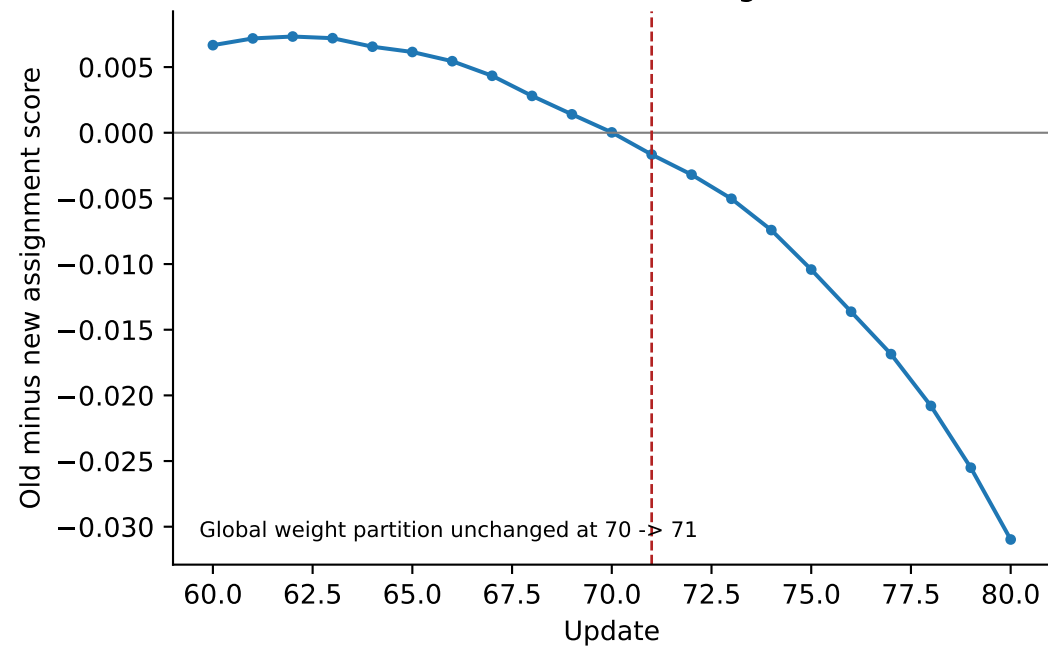


# Crossing evidence: controlled construction, natural trajectory, local Euclidean reproduction

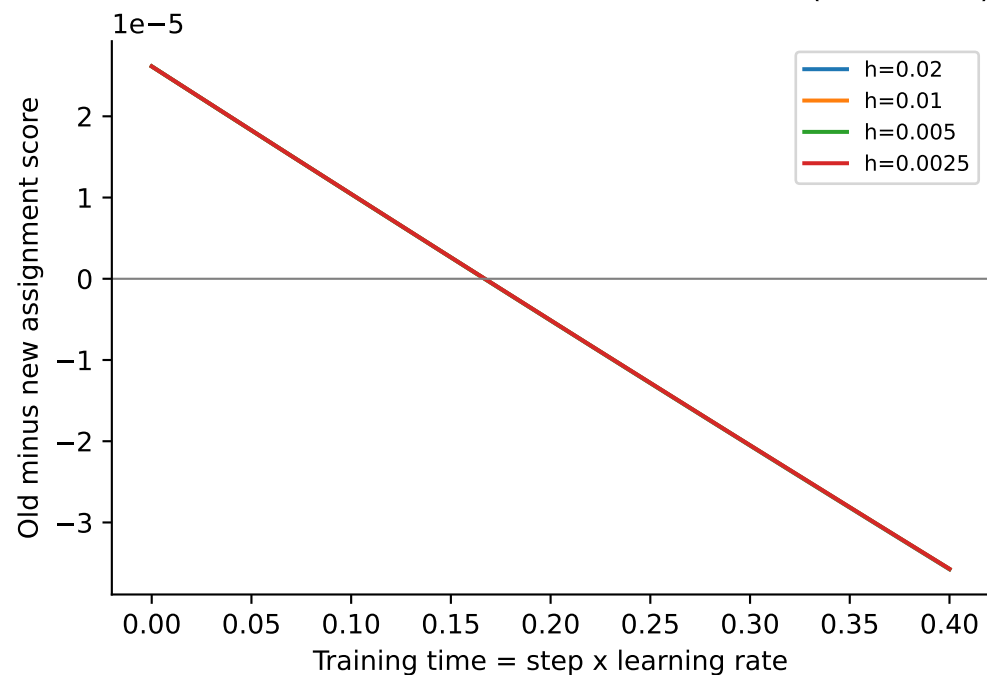
## A. Controlled fixed quadratic loss



## B. Natural SmolLM2 training (AdamW)



## C. Same real trained state + fixed real text (FP64 SGD)



## D. Real LM: numerical refinement

